Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Machine Learning Engineer with 3 years of industry experience building and operating production **Python** systems, including **self-hosted Llama-3 inference on a GPU EC2 instance**, where I made the quality-versus-cost trade-off explicit for the largest department. Built end-to-end **RAG and retrieval** services over **PostgreSQL/pgvector** and the **AWS** infrastructure behind them (EC2 including a GPU instance, S3, ECR, Lambda, Docker, one-click Jenkins and GitLab CI/CD). On an MSc in Data Science (University of Bristol, awaiting result), trained small CNN/ViT encoders and sequence decoders from scratch and evaluated encoder-decoder architectures across multiple seeds with SBERT cosine, BLEU and ROUGE-L; co-built an evidence-backed scoring and ranking tool with **AstraZeneca**. Acted as de facto team lead for up to 4 interns and was twice recognised with **Best Team Project** (2024, 2025). Seeking to work on ML and HPC architecture, model inference and hardware evaluation at G-Research.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Machine Learning & Retrieval Systems

- Designed and built production **GenAI** applications end to end (**ReactJS** frontend, **Python FastAPI** microservices, **RAG** with **LlamaIndex** and **Azure OpenAI** GPT-3.5 & **Llama-3**) serving Compliance, Client Service and Investment, with **Azure OpenAI embeddings** and a separate **pgvector** database per department.
- Built **tool-calling** agents that call **Python** functions to run **SQL**-backed fund comparisons and generate PDF reports on request; guardrailed behaviour with **system prompts**, versioned every prompt change in **Git** and had the **Compliance** department review each release.

GPU Inference, Compute & Cost

- Chose and operated a per-department model mix on quality and cost: **self-hosted Llama-3** on a **GPU EC2** instance for the largest department's traffic, with the stronger **Azure OpenAI** outputs kept for decision makers; released to real users and changed the applications based on what they found.
- Designed the infrastructure for internal analytics platforms and GenAI applications on **AWS** (**EC2** including a **GPU** instance, **S3**, **ECR** and **Lambda**); packaged the **FastAPI** microservices (**uvicorn**) as **Docker** containers and automated deployment of the separate frontend and backend repositories (dev and prod) through a one-click **Jenkins** and **GitLab CI/CD** pipeline.
- Restricted access to company private IPs using **AWS Security Groups**, implemented **RBAC** with a **Python** authentication function against the company **LDAP** server and deployed an **Nginx** reverse proxy, one of the fixes that took the applications from repeatedly failing to passing the security team's **VAPT**.

Data Pipelines & Stakeholder Delivery

- Built a **Python** pipeline that parses over **300 daily emails** from credit rating agencies into structured data, fuzzy-matches company names against the ones analysts track and sends a daily summary, saving **60 analyst hours a month**; parsed first with **AWS Textract**, then with **Azure OpenAI** when a rating agency began sending emails as images.

- Designed a Normalized SQL Schema (**PostgreSQL**) and wrote **Python** ingestion scripts for the **Morningstar** API covering every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly and monthly and integrated with legacy analyst data; developed **Optimized SQL** queries, Functions and Procedures.
- Migrated financial analysts from Excel to **Power BI** dashboards, improving efficiency by **10 hours a week**; analyzed customer data using Python **Pandas** and Market Basket Analysis (Apriori Algorithm).
- Worked directly with stakeholders in Client Service, Compliance and Investment to turn requirements into working tools; acted as **de facto team lead** for up to **4 interns** in my final year; recognised with **Best Team Project** twice (2024 rating-email automation, 2025 department RAG chatbot).

PROJECTS

- University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca: CellLineSelector (**Team project, co-built**. Preprocessing pipeline with identity resolution and integrity checks, plus a **scoring and ranking** algorithm that ranks cancer cell lines by target inclusion and exclusion genes using per-gene scores across six evidence layers, calibrated across lineages, and returns the evidence behind each pick; validated and refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- University of Bristol MSc Applied Data Science Group Project: VisionCaptioner (**Team project**. Trained SimpleCNN, ResNet-18 and a custom ViT encoder from scratch and compared symbolic, SBERT embedding-retrieval and LSTM/attention decoders for generating image descriptions on three synthetic datasets (10,000 images each), across **five seeds** with Adam, and evaluated them with **SBERT cosine similarity, BLEU and ROUGE-L** against BLIP/SigLIP and commercial VLMs.) GitHub: https://github.com/Arishtanemi4/ata-team30.git
- Wingman: AI dating-coach web app with 108 LLM personas across 12 archetypes and a vision model scoring uploaded photos (ReactJS, Python FastAPI, Kimi K2 and DeepSeek V4 Pro via NVIDIA NIM). LLM output treated as untrusted: validated with Pydantic, repaired when malformed and cached by input hash; rate limits shaped the design (12 calls instead of 108, serialised calls, results streamed to the UI). GitHub: https://github.com/Arishtanemi4/wingman.git

ACHIEVEMENTS

- Project Hack 27: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- Encode Vibe Coding Hackathon: Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- University of Bristol Datathon sponsored by Lloyd's Bank: 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- Tools: Docker, Jenkins, GitLab (CI/CD), AWS (EC2 including a GPU instance, S3, ECR, Lambda, EventBridge, SageMaker, Textract), Azure OpenAI, LlamaIndex, Nginx, Linux, LDAP, Power BI, VS Code, Claude Code.
- Languages: Python (pandas, scikit-learn, FastAPI, Pydantic, uvicorn), SQL, R (dplyr, tidyverse, ggplot2), NodeJS (express, ReactJS), Shell
- Databases: PostgreSQL (pgvector semantic search, SQL functions and procedures), SQLite